Superconductivity

The measured table, the missing math, and the coherence route to room temperature.

v0.2 · 2026-08-01 (drafted, scored, and consolidated same day) · status-labelled throughout: MEASURED published third-party measurement · DERIVED bookkeeping from measured numbers · PRE-REG frozen prediction / frozen gate · AUDIT lore positioned against budget, no reality claim · OPEN honest unknown · DISPROVEN refuted or retracted. Firewall: research-lane document; not deck material.

0 · What this paper is

Four jobs. **One**: the complete measured record — every major superconductor family, its critical temperature, its pressure, and whether its mechanism is solved. The record's punchline is not widely understood: *the highest-T_c superconductors ever confirmed are the best-understood ones*, predicted from first principles before they were synthesized. The math famine is real but local — it covers the correlated-electron families, not superconductivity itself. **Two**: the mechanism story from first principles — starting from what an electron actually is, since pairing is a statement about the electron's collective vacuum, not about gluing two marbles together. **Three**: the routes to room temperature — what the solved half already prescribes (§6.1's 300 K convergence and double screen), the driven/ nonequilibrium route (§5, with the tuned-vs-stirred discriminator of §5.2), the intuition candidate (§6.3), and a staged bench program for a duty-cycled room-temperature superconducting state (§6.4). **Four**: the adjudications — the Pais patent + AIAA mechanism (audit linked in banner; its bench doubles as a replication of the orphaned Gupta 2010 Al/PZT anomaly, §5.1) and the Schön fulleride scandal (fabrication, not suppression; the "no measured λ, no promotion" rule in §2.1 is its residue). Status at v0.2: the §7 pre-registration was *scored the same day it was frozen* — P1 PASS, P3 PASS, P4a PASS (nine materials cluster at $\bar{r}$ = 0.607 of the consensus ceiling; the 1/φ identification remains indeterminate), P2 recorded as a negative. Every OPH-specific claim is tagged OPEN, PRE-REG, or scoreboarded DERIVED, per the same discipline as Sophia.

1 · What is an electron, from first principles

1.1 The free electron — the most precisely measured object in science

MEASURED The electron is an excitation of a quantum field: mass 510.998 950 69(16) keV/c², charge −e exactly (charge quantization, Millikan 1913 onward), spin ½. Its magnetic moment matches the QED prediction to about one part in 10^{12} (Fan, Myers, Sukra & Gabrielse 2023) — the most precisely verified prediction in the history of science. Collider contact-interaction bounds put any substructure below $\sim 10^{-18}$ m; the electric-dipole-moment bound ($|d_e| < 4.1\times10^{-30}$ e·cm, JILA 2023) makes it the roundest object ever measured. Spin ½ means the electron lives on the *double cover* of the rotation group: a 2π rotation flips the wavefunction's sign — not philosophy, measured directly in neutron interferometry (Rauch 1975) and implicit in every atom.

1.2 The electron in matter — a quasiparticle, and it can shatter

MEASURED Inside a solid, "the electron" is not the bare field excitation. It is a *quasiparticle* — the bare electron dressed in the response of every other electron and the lattice (Landau Fermi-liquid theory). The dressing is not small: effective masses in heavy-fermion metals reach $\sim 1000\ m_e$. And the dressed object is not indivisible: in one-dimensional conductors the electron measurably *fractionalizes* into separate spin and charge excitations (spinon/holon separation: Kim et al. ARPES 2006, Jompol et al. *Science* 2009), and in the fractional quantum Hall regime carriers of charge e/3 have been directly detected in shot noise (Saminadayar et al., de-Picciotto et al., 1997). The lesson that matters for this paper: **what conducts in a material is whatever collective excitation the many-body ground state supports**. Superconductivity is the extreme case — the ground state reorganizes so that the charge-carrying excitation is a *pair condensate* with a single shared phase.

MEASURED That the carriers pair is not an interpretation: magnetic flux through a superconducting ring is quantized in units of h/2e — the "2" is Cooper pairing weighed directly (Deaver–Fairbank and Doll–Näbauer, 1961). That the condensate has one macroscopic phase is also not an interpretation: the Josephson effect turns that phase into a frequency so exactly that the SI volt is *defined* through it.

1.3 The OPH dialect

OPEN In the programme's vocabulary (meta-frame), the electron is a patch/port excitation. The precise status of the dialect, checked against the network (2026-08-01): **(a)** the double cover behind spin has a named *geometric candidate* — the Shape paper's eversion passage, where the halfway state of the sphere turning inside out carries exactly the one-turn-swaps-sides / two-turns-identity symmetry — and the 2026-07-06 sim lane reproduced that *phase* double cover. **(b)** Exclusion/occupancy — the statistics rule, two per orbital, fill lowest first — remains *assumed, not derived*, anywhere in the network. The sharpest evidence is the S5 closure itself: the Periodic Table paper's TF-as-consensus run derives the aufbau *ordering* (6/6, zero fits) while *importing* occupancy through Thomas–Fermi statistics — the $n \propto p_F^3$ closure *is* "fill phase space two per cell," handed in as input. That the closure landed exactly when ordering was asked of consensus and occupancy was supplied by hand confirms the 2026-07-06 three-ingredient split (degeneracy: derived; ordering: derived; occupancy: statistics rule, still open — the entanglement-wall side). Nothing later in this paper leans on §1.3 except the pre-registered §7; if §7's predictions fail, this subsection stays what it is — a dialect, not a derivation.

2 · The measured table

MEASURED Every family that has cleared third-party replication. T_c at ambient pressure unless noted.

Material / family	Year	T_c (K)	Pressure	Class	Mechanism status
Hg (Onnes — the discovery)	1911	4.2	ambient	elemental	solved (BCS)
Pb	1913	7.2	ambient	elemental	solved (strong-coupling Eliashberg)
Nb	1930	9.3	ambient	elemental	solved
NbTi (every MRI magnet)	1962	9.8	ambient	alloy	solved

Nb_3Sn / Nb_3Ge (A15)	1954/73	18.3 / 23.2	ambient	A15	solved; record-holder until 1986
MgB_2	2001	39	ambient	covalent metal	solved (two-gap, phonon; ab initio reproduced)
K_3C_{60} → Cs_3C_{60}	1991/2008	19 → 38	Cs_3C_{60} under ~kbar	fulleride	mostly solved (phonon + correlation)
$(TMTSF)_2PF_6$, κ-BEDT salts	1980+	~1 / ~12	some under pressure	organic	open (unconventional)
$CeCu_2Si_2$, UPt_3, UTe_2	1979+	0.5–2	ambient	heavy fermion	open (magnetically mediated; UTe_2 triplet candidate)
$La_{2-x}Ba_xCuO_4$ (Bednorz–Müller)	1986	35	ambient	cuprate	**open**
$YBa_2Cu_3O_7$ (first past LN_2)	1987	92	ambient	cuprate	**open**
Bi-2223 / Tl-2223	1988	110 / 125	ambient	cuprate	**open**
$HgBa_2Ca_2Cu_3O_8$ (ambient record)	1993	133 (~164 @ 30 GPa)	ambient / high-P	cuprate	**open**
Sr_2RuO_4	1994	1.5	ambient	ruthenate	open (order parameter still contested)
LaFeAsO:F (Hosono) → SmFeAsO:F	2008	26 → 55	ambient	iron pnictide	**open** (spin-fluctuation s±, likely)
FeSe bulk → FeSe/$SrTiO_3$ monolayer	2008/12	8 → ~65 (gap)	ambient	iron chalcogenide	open; interface enhancement ×8 measured

Magic-angle bilayer graphene	2018	~1.7	ambient	moiré	open; fully gate-tunable testbed
$Nd_{0.8}Sr_{0.2}NiO_2$ (infinite-layer)	2019	9–15	ambient (thin film)	nickelate	open
$La_3Ni_2O_7$	2023	~80	~15–20 GPa	nickelate	open; ambient-pressure film reports (~40 K) still stabilizing
H_3S (predicted → then found)	2015	**203**	~155 GPa	hydride	**solved** (Eliashberg; predicted Duan et al. 2014)
LaH_{10} (predicted → then found)	2019	**~250–260**	~170–190 GPa	hydride	**solved** (predicted Liu–Hemley & Peng 2017)
CaH_6 / YH_9	2021–22	215 / 243	~170–200 GPa	hydride	solved (same machinery)

2.1 The cautionary sub-table

DISPROVEN Logged because this field attracts spectacular claims, and the project's hygiene requires the failures on the same page as the successes:

Claim	Year	Outcome
Schön (Bell Labs) organic T_c series	2001–02	Fabricated; papers withdrawn.
Dias CSH "288 K @ 267 GPa"	2020	Retracted 2022 (data provenance).
Dias Lu–N–H "294 K near-ambient"	2023	Retracted 2023; investigation found fabrication.
LK-99 "room-T ambient-P"	2023	Replications: Cu_2S structural transition at 104 °C + ferromagnetic shards; not a superconductor.

Pattern worth naming: every failure was a *resistance-drop-without- Meissner-thermodynamics* story. The receipts that cannot be faked together are: zero resistance *and* flux expulsion *and* the specific-heat anomaly at the same T_c. Any single one alone has a mundane imposter.

AUDIT The Schön case got a full adjudication (operator question: fraud vs suppression vs failed replication) in the Schön fulleride audit. Verdict in one line: the evidence was *internal* (identical noise curves across different "experiments"), incentives ran pro-discovery, and the adjacent real physics (fulleride SC, gate-induced SC via ionic liquids, driven K_3C_{60}) thrived rather than being suppressed — fabrication, not suppression. Two durable lessons folded back into this paper: the stiffness screen is *necessary-not-sufficient* (Schön's 117 K claim computes to a plausible r ≈ 0.43 — fakes are crafted inside the allowed envelope), and the fraud pattern is transport-first claims with the stiffness observable missing. Practical promotion rule: **no measured λ, no promotion.**

3 · The two halves of the problem

3.1 The solved half — and how completely it is solved

MEASURED For phonon-mediated superconductors the theory chain BCS (1957) → Eliashberg (1960) → density-functional superconductivity is not merely explanatory but *predictive with no free parameters*. The receipts, in ascending order of impressiveness: the isotope effect ($T_c \propto M^{-1/2}$, measured 1950, the clue that pointed at phonons before BCS); the universal gap ratio $2\Delta/k_BT_c \approx 3.5$ across the conventional table; MgB_2's two-gap structure reproduced ab initio; and the capstone — **H_3S and LaH_{10} were computed first and synthesized second**. Theorists solved the electron–phonon problem for compressed hydrides, published $T_c \approx 200$ K numbers, and the diamond-anvil experiments then found superconductivity within tens of kelvin of the predictions. The strongest superconductors known are the ones the math owns outright.

3.2 The open half — precisely where the math stops

MEASURED The correlated families (cuprates above all) have resisted 40 years of attack for a reason that is itself sharp: the minimal model believed to contain them — the 2D Hubbard model, two terms, written down in 1963 — cannot be solved. There is no small parameter; quantum Monte Carlo is blocked by the fermion sign problem, which is NP-hard in general (Troyer–Wiese 2005); and the best modern controlled numerics (the Simons collaboration's cross-validated DMRG + AFQMC campaigns) find stripe order and d-wave pairing *competing within tenths of a percent of the ground-state energy*. Four decades in, it is not settled whether the minimal model even superconducts at cuprate temperatures. No ab initio T_c prediction has ever landed for a cuprate. That — not superconductivity in general — is the famine.

3.3 What the whole table obeys anyway

DERIVED Even without the correlated mechanism, the table has laws:

- **McMillan / Allen–Dynes**: T_c rises with (phonon frequency scale) × (coupling strength). This single law explains the entire hydride program: hydrogen is the lightest atom, so metallic hydrogen-rich lattices have the highest phonon frequencies available to chemistry — megabar pressure is just the price of metallizing them.
- **Uemura relation**: in the unconventional families, T_c tracks the *superfluid phase stiffness* (condensate density / mass), not the pairing scale — coherence, not glue, is the bottleneck.
- **The dome**: unconventional T_c versus doping is a dome, with the strange-metal region above its summit — universal across cuprates, pnictides, organics, moiré systems.
- **Proximity to magnetism**: essentially every unconventional family sits next to an antiferromagnetic (or spin-density) phase; superconductivity appears where magnetic order melts. Spin fluctuations as the glue is the leading (unproven) suspect across all of them.

4 · Coherence is the mechanism: superconductivity as laminar charge flow

MEASURED The state itself, as opposed to the pairing glue, is understood — and it is a fluid statement. The Madelung transform maps the condensate wavefunction exactly onto a fluid: phase gradient = velocity field. A superconductor is a charged fluid whose viscosity-free ("laminar") flow is protected by two facts: the energy gap removes the soft excitations that dissipation needs, and the single macroscopic phase makes vorticity *quantized* — the fluid cannot develop smooth eddies, only discrete flux lines. Dissipation in a superconductor is, measurably, vortex motion: type-II resistance is flux-flow drag; the 2D transition (BKT) is literally vortex unbinding. In project vocabulary: **the condensate is a phase consensus; supercurrent is what flow looks like while consensus holds; resistance is consensus breakdown by vortex proliferation** — the same failure mode as turbulence onset in the neutral fluid, with the gap playing the role viscosity cannot.

> **Meissner causality box.** Flux expulsion is a property *of* the superconducting state, not a road into it. Three receipts, two of them arrow-of-time-free: (1) *double dissociation* — zero-field-cooled superconductors superconduct with nothing to expel, while externally imposed flux-free conditions never make a normal metal superconduct; (2) *type-II coexistence* — between H_{c1} and H_{c2} flux penetrates as vortices and superconductivity persists (every NbTi magnet operates there); (3) *energetics* — expulsion costs $B^2/2\mu_0$, paid from condensation energy, verified quantitatively via the Rutgers/critical-field thermodynamics. Receipt (3) assumes relaxation toward equilibrium; (1) and (2) are observed facts with no temporal-ordering assumption. Perfect diamagnetism *is* the defining signature of the state (Anderson–Higgs: the photon gains mass inside) — definition-grade signature, not drivable cause.

5 · The driven route — measured, and its impostor

MEASURED Coherently pumping one targeted lattice mode can push superconducting-like transport above the equilibrium T_c: Fausti et al. (*Science* 2011, stripe-ordered cuprate, resonant THz phonon drive), Mitrano et al. (*Nature* 2016, K_3C_{60}, superconducting-like optical response far above T_c), Budden et al. (*Nat. Phys.* 2021, K_3C_{60} metastable for >10 ns). The mechanism family is nonlinear phononics: drive one infrared-active mode hard; anharmonic coupling displaces the lattice along a direction that strengthens pairing. Two disciplines to carry out of this: it is **coherent drive of a specific mode** — an incoherent phonon bath is just heat, and heat kills the consensus; and the operative band is **THz**, because that is where optical phonons live.

AUDIT The Pais patent (US20190348597A1, abandoned) claims the same family outcome — vibration- induced room-temperature superconductivity — via macroscopic piezoelectric drive. Full adjudication in the audit; summary: the stated mechanism fails a kinematic bound. A structure's maximum mechanical ring rate is `f ≈ v_sound/L`; a millimeter wire caps near 5 MHz, and reaching the THz band that pairing physics responds to requires features at the lattice constant itself — at which point the drive that works is light, i.e. the measured Cavalleri route. This bound is a property of the wave equation and holds identically under time reversal; the arrow-of-time objection (operator, 2026-08-01) applies only to the free-energy leg of the Meissner box above, and that leg is corroborated in-project by the 2026-07-06 physics-fidelity audit's measured emergent arrow. The audit killed the patent's *stated mechanism*, not every possible behavior of the artifact — which is what §5.1 is for.

5.1 Pre-registered bench test of the Pais wire

PRE-REG Frozen before any build, so the outcome is a finding either way. **Device**: per the patent — insulated core wire, piezoelectric coating, pulsed current drive, applied magnetic field, room temperature. **H1**: four-wire resistance under drive drops by ≥50% relative to the same wire undriven (threshold frozen; a true SC transition is orders beyond this, so the gate is generous). **Controls**: (C1) same wire, drive off; (C2) same drive waveform into a geometrically matched non-piezo-coated wire; (C3) thermal control — resistance logged against wire temperature to catch simple heating-artifact signs (resistance *rise* masking as anomaly, or contact artifacts). **Falsification**: H1 fails → the audit's verdict is upgraded from computed prior to receipt-grade negative, and the patent's mechanism is closed in this project. H1 passes with controls clean → drop everything and replicate. Predicted outcome, stated for the record: null, confidence >99%. That number is exactly what pre-registration exists to discipline.

AUDIT *[2026-08-01, after reading Pais's AIAA companion paper (10.2514/6.2019-0869) in full — audit Addendum 3.]* Two updates. The paper's mechanism chain fails at one identifiable step: Pais himself concedes the lattice must reach $\omega \gtrsim 10^{12}$ Hz ("extremely difficult to obtain by physically vibrating the wire" — our bandwidth objection, his words) and bridges the gap with $\omega_c \sim \omega^2 \cdot t_{op}$, "frequency accumulating with operating time" — refuted by every quartz watch (32 kHz × a year of t_{op} would emit beyond-THz radiation). And the design's provenance surfaced: his wire is built around Gupta, arXiv:1007.2736 (2010) — an unreplicated, λ-free report of resistance anomalies at ~313 K at an Al/PZT interface. **This bench is therefore also a replication attempt of Gupta 2010**, which raises its value while leaving the predicted null unchanged; control C3 does the work the original report lacked.

5.2 Drive as dynamical p-tuning — tuned, not stirred

OPEN The operator's mapping, adopted with one sharpening: coherent drive is a *fifth coordinate reading of p-tuning*. The programme's p-tuning knob has four static readings (geometric aspect, modal phase closure, curvature/tension, repair exponent); Floquet engineering — the mainstream name for what a coherent periodic drive does — *renormalizes a material's effective couplings while the drive is on*, i.e. it moves the same dial dynamically that chemistry and geometry move statically. Under this reading, the §5 beat-note program is dynamic p-tuning applied at a phonon port, and Pais's "driving non-equilibrium dynamics is essential" names the right *family*.

MEASURED The discriminator that splits the family — **tuned beats stirred, everywhere it has been measured**: (a) In K_3C_{60} the pump must be resonant with a specific intramolecular phonon; detune off the mode and the effect dies — it is not "non-equilibrium" that is essential, it is *which mode*. (b) Dynamical decoupling (spin echo → CPMG) genuinely suppresses decoherence with periodic drive — but only with *phase-coherent* pulse sequences; random kicks decohere faster. (c) Floquet prethermalization gives driven systems exponentially slow heating — but only for high-frequency, narrow-band drive above a spectral gap. (d) Discrete time crystals (trapped ions 2017 →) are literally "never-equilibrating driven coherent order" — Pais's sentence, as a measured phenomenon — and every realization needs coherent periodic drive PLUS a heating bottleneck (localization or a prethermal gap). (e) Room-temperature polariton condensates (organic microcavities, 2013–14) are macroscopic quantum coherence sustained *by* drive far from equilibrium — in a cavity-protected mode. Five instances, one grammar: **coherent narrow-band drive + a protection mechanism → long-lived driven order; broadband "abrupt" agitation → heat.** Pais's prescription ("soup of fluctuations," Prigogine order-from-chaos) selects the stirred member of the family, which is the one that measurably fails; Prigogine's dissipative structures are real but classical — no theorem connects delayed entropy production to *quantum phase* coherence, and that unbridged gap is where the paragraph jumps. The honest completion of his argument is: never let the system thermalize, *by driving one protected mode coherently* — which is Cavalleri's program, our §5, and the tuned-detuning doctrine (φ-bulk + engineered ports) in its dynamical form.

6 · The routes to room temperature at ambient pressure

DERIVED What the solved half already prescribes: maximize (phonon scale × coupling) without megabar pressure. The live directions, ranked by receipt strength: **(a)** ternary/clathrate hydrides engineered for chemical pre-compression (the La–B–H / Y–B–H programs: keep LaH_{10}'s hydrogen cage, replace external GPa with internal lattice pressure); **(b)** metastable retention — synthesize under pressure, quench to ambient, as diamond itself demonstrates is possible in principle; **(c)** covalent-metal design in the MgB_2 direction (light-element σ-bond metals); **(d)** interface engineering — the measured ×8 enhancement of FeSe on $SrTiO_3$ shows a substrate phonon can multiply a T_c; nothing forbids stacking that trick. OPEN What the unsolved half would unlock: the cuprates already run at 133 K at ambient pressure *without our understanding them*. A correct theory of the correlated class is the one credible path to *designed* ambient-pressure room-temperature superconductivity, which is why the Hubbard problem — or a consensus-dynamics reformulation of it — is the highest-value open target in this paper.

6.1 The 300 K convergence — the two halves merge at room temperature

DERIVED The P1/P4 scoring adds a constraint the glue-only literature underweights: at 300 K the solved half's route and the open half's bound *converge*. Any room-temperature superconductor needs $T_\theta \gtrsim 300$ K — $\lambda \lesssim \sim 120$ nm at cuprate-like layer spacing — and the strong-coupling glue class is already drifting toward that ceiling as T_c climbs (H_3S at r = 0.167; the classic BCS metals sit at 10^{-8}–10^{-3}). Extrapolating the trend, a 300 K material lands at r ≳ 0.3–0.5 *regardless of mechanism* — i.e., inside the band where the unconventional class lives today. Corollary, stated as a falsifiable expectation: **any genuine ambient room-temperature superconductor will phenomenologically resemble a cuprate** — wide fluctuation regime, BKT-flavored transition, vortex-liquid problems, strong sensitivity to disorder — even if its pairing is plain Allen–Dynes phonon glue. "Room-temperature but conventional-behaving" claims should be treated as self-undermining under this screen.

DERIVED Candidate shortlist under *both* screens (glue: high $\omega \times \lambda_{ep}$; stiffness: $\lambda \lesssim 120$ nm — and per §2.1, no measured λ, no promotion): **(a)** ambient-pressure ternary hydrides from the ab initio pipeline — the standout being Mg_2IrH_6 (predicted T_c ~80–160 K at ambient pressure; *unsynthesized/unverified*, flagged as such) and the $LaBeH_8$-class low-pressure clathrates; **(b)** light-element covalent metals in the MgB_2

lineage (B–C–H phases, heavily doped diamond as the proven-in-principle carbon endpoint); **(c)** interface multiplication ($FeSe/SrTiO_3$'s measured ×8) stacked on either. The carbon family's own record is instructive: pristine graphene and nanotubes do not superconduct (vanishing carrier density — zero stiffness), and every receipted route in is a *detuning*: chemical (Ca-intercalated graphite/graphene, 11.5 K), geometric (magic-angle twist, §6.2), curvature+doping (CNT reports at 0.5–15 K, 1D-fluctuation-fragile and partially contested).

6.2 Magic-angle graphene — the detuning archetype, and the ceiling made visible

MEASURED Twisted bilayer graphene is the cleanest existing embodiment of the protected-bulk-plus-tuned-leak pattern: two *perfect* lattices, aligned exactly, are an ordinary semimetal — frozen; rotate one by the specific small detuning of ~1.1° (the magic angle) and moiré flat bands appear, hosting superconducting domes adjacent to correlated-insulator states, all gate-tunable in one device. Note the two-sided lesson: the flat band that enables pairing also makes carriers heavy, which *crushes phase stiffness* — TBG's T_c (~1.7 K) is measurably stiffness-throttled, the P1 bottleneck in miniature. Detuning opens the channel; the ceiling then prices it. A designed RT-SC must win both sides at once, which is exactly what §6.1's double screen encodes.

6.3 The intuition candidate — ϕ-frustrated clathrate hydride

OPEN Operator-requested: the best single guess, assembled from this paper's receipts, register marked honestly as intuition. Why no perfect RT-SC exists yet, stated as trade-offs: strong coupling wants flat bands, which crush stiffness (the TBG lesson); high phonon scales want light atoms, which demand pressure to metallize (the hydride lesson); high DOS near an instability invites a competing striation order that steals the ground state (the cuprate lesson). A room-temperature superconductor is not a single-knob extremum — it must sit at the tuned point of several consensus channels at once. The guess, one ingredient per constraint:

- **Cage: icosahedral boron/carbon framework.** Covalent, light, stiff — MgB_2-class σ-band metallicity and high phonon scales; boron's native chemistry is already icosahedral clusters (β-boron). Supplies stiffness (dense light carriers → small λ) and glue backbone.

- **Filling: hydrogen in the cage voids.** Chemical pre-compression — the LaH_{10} mechanism at ambient pressure; hydrogen modes extend the phonon spectrum to the highest frequencies chemistry allows.
- **Global structure: quasicrystalline / Fibonacci-layered (φ-aperiodic).** The load-bearing new idea, and where the project's KAM thread earns a falsifiable job: stripe/CDW/SDW rivals are *commensuration lock-ins* — rational-ratio resonances between carrier density and lattice period. A φ-aperiodic host frustrates commensurate locking the same way a golden winding frustrates mode-locking — while uniform phase consensus (superconductivity), which requires no commensuration, is untouched. Receipt-adjacent anchors: quasicrystal superconductivity exists (Al-Zn-Mg, 2018 — compatibility proven, T_c tiny because glue is weak there); stripe order demands commensuration (§3.2's Hubbard result). Aperiodicity kills the rival, not the condensate.
- **Carrier tuning: dope to the operating band.** P4's receipt — optimized superconductors sit at r ≈ 0.6 of their stiffness ceiling; the electron count is the detuning knob that puts the material there.
- **Last-mile option: beat-note phonon drive (§5)** on a near-miss composition — coherent THz difference-frequency pumping as the bridge from a 200 K material to a 300 K demonstration, per the measured K_3C_{60} metastability lineage.

Falsification path (the honest exit): any synthesized candidate must clear the §6.1 double screen — measured λ ≤ ~120 nm *and* phonon spectrum reaching multi-THz with strong coupling — and per §2.1, no measured λ, no promotion. Every ingredient above has a receipt; the combination has none. That is what "intuition, tagged OPEN" means.

6.4 The bench program — a duty-cycled room-temperature superconductor

DERIVED The buildable object is not the equilibrium material (§6.3, a synthesis program) but the **driven metastable state**, staged on three published receipts: YBCO pumped at its ~17–20 THz apical-oxygen phonon shows transient superconducting-like c-axis response *up to room temperature* (Hu/Kaiser/Cavalleri 2014); K_3C_{60} shows the same family with a *metastable* state living >10 ns after a ~1 ps pump (Budden 2021); and resonant tuning cuts the required fluence ~100× (Rowe 2023). The program:

- **Stage 0 — equilibrium baseline (~$500).** Commercial YBCO coated tape + LN_2: four-probe transition at 92 K, two-coil mutual-inductance screening, magnet levitation. Calibrates every probe used later on a real superconductor. No exotic parts.
- **Stage 1 — driven state above equilibrium T_c, cryo-assisted.** Same YBCO tape at 100–200 K, pumped in the mid-IR at the apical-oxygen resonance; ns-resolved mutual-inductance + microwave reflection looking for drive-synchronous screening. **The core control is tuned-vs-stirred (§5.2): on-resonance vs off-resonance pump at equal absorbed power.** A thermal or trivial artifact follows absorbed power; the claimed mechanism follows the phonon resonance. This one control discriminates everything.
- **Stage 2 — the 300 K shot.** Identical protocol at room temperature (replicating the 2014 YBCO geometry with independent probes). Gate: drive-synchronous diamagnetic screening at 300 K, on-resonance only, scaling with fluence, absent in the detuned control.
- **Stage 3 — duty-cycle sustain.** Exploit metastability: if the driven state lives τ after each pump and the source repeats every $t < \tau$, the state is quasi-continuous. Budden's 10 ns at 100 MHz rep rate is already unity duty cycle in principle; the stage-3 frontier is lifetime engineering at 300 K.

OPEN The honest cost wall and the project's angle on it: published drives are fs-OPA-class fields (~MV/cm peak — $100k+ instruments); CW sources (QCLs, DFG beat-notes from detuned carriers per §5) sit orders of magnitude below in peak field even with the resonant ×100 discount. The gap is a *field-enhancement* problem, which is a resonator problem — the project's home turf: a high-Q THz microcavity at the phonon frequency (φ-detuned body, port- engineered, film at the antinode) recovering 10^2–10^3 in field is the chamber doctrine applied to this bench, and the difference between "replication with borrowed laser" and "standing tabletop instrument." Promotion discipline unchanged: transient optical signatures alone are contested territory in the literature — our gate is drive-synchronous *flux exclusion* plus transport, and for any equilibrium claim the full §2.1 trio. No measured λ, no promotion.

7 · OPH pre-registration — frozen before scoring

[Scoreboard update, 2026-08-01 — scored same day, gates frozen first in `lpoh/sims/PREREG_SC_P1P3_2026-08-01.md`*.]* DERIVED **P1 PASS · P3 PASS · P2 FAIL as frozen.** The consensus sim (2D XY phase reconciliation) produced its universal constant c = 0.929 (gate: [0.80, 1.00]; infinite-L literature 0.893), and the resulting zero-fit stiffness ceiling $k_B T_\theta = c\cdot(\hbar^2/4)\cdot L_{eff}/(\mu_0 e^2 \lambda^2)$ splits the frozen material table exactly as predicted: the unconventional four (LSCO, YBCO, Bi-2212, BaK-122) sit at **0.40–0.81 of the consensus ceiling** (stiffness-bound, Uemura-form slope 1.11 ± gate), the conventional five (Al→MgB_2) at 10^{-8}–6×10^{-3} (glue-bound), separation 63× with no overlap. P2's sim gate missed (fluctuation peak 0.24 vs endpoint marker 0.36; candidate criterion artifact, recorded as a negative — any retest is a new pre-reg). Unplanned finding: H_3S sits at **0.167** of its ceiling — the strong-coupling hydrides are measurably drifting toward the stiffness limit, which yields a falsifiable screen for any future room-temperature claim: ambient RT-SC needs $T_\theta \gtrsim 300$ K, i.e. $\lambda \lesssim 120$ nm at cuprate-like layer spacing. Receipts: `lpoh/sims/RESULTS_SC_P1P3_2026-08-01.md` + `SC_p1p3_consensus.py` + `SC_p1p3_results.json`.

[Scoreboard update 2, same day — P4, the "detuning constant" test, frozen in `lpoh/sims/PREREG_SC_P4_2026-08-01.md` *before scoring.]* DERIVED **P4a PASS · P4b indeterminate · P4c PASS.** Expanding to nine unconventional materials across three families: the ceiling ratios **cluster** (std(ln r) = 0.358, gate ≤ 0.45) around geometric mean $\bar{r}$ **= 0.607**, 95% CI [0.485, 0.770] — a universal operating fraction of the consensus ceiling is now a receipted pattern. The pre-flagged NUMEROLOGY-RISK identification $\bar{r} = 1/\varphi = 0.618$ survives the CI but so do the rivals 0.50 and 0.75: **clustered, value-indeterminate** as frozen — discriminating needs tighter λ data under a new prereg. Bonus receipts: underdoped YBCO computes to r = 1.20 ± 0.3 — the framework locates, with no fit, exactly the corner of the phase diagram the literature independently calls phase-fluctuation-limited; and the dome is bracketed (underdoped ≈ 1.2 → optimal ≈ 0.6 → overdoped ≈ 0.8), so the ~0.6 clustering is a property of *optimized* compounds — empirical materials optimization drives superconductors toward a common detuning from their own consensus ceiling. Receipts: `lpoh/sims/RESULTS_SC_P4_2026-08-01.md` + `SC_p4_ratio_cluster.py` + `SC_p4_results.json`.

PRE-REG The consensus framework enters here or not at all; zero results exist at freeze time. Precedent: the S5 TF-consensus result closed the aufbau ordering with zero fitted parameters — the same shape is demanded here. Frozen targets:

- **P1 (stiffness law)**: derive, from consensus dynamics with no fitted parameters, that unconventional T_c is set by phase stiffness rather than pairing scale — i.e., reproduce the *form* of the Uemura relation. Score: does the derived exponent/proportionality match the measured Uemura plot across ≥3 families?
- **P2 (competition ordering)**: from the same dynamics, predict the ordering "superconductivity appears where magnetic consensus melts" — concretely, that every unconventional dome sits adjacent to a magnetically ordered phase in the phase diagram, and predict on which side the dome's maximum sits relative to the magnetic endpoint. Score against the measured phase diagrams of cuprates, pnictides, heavy fermions, κ-organics.
- **P3 (negative control)**: the framework must NOT predict a stiffness bottleneck for the conventional/hydride class (where T_c tracks the Allen–Dynes glue scale instead). A framework that "explains" both halves the same way explains neither.

Falsification: P1 or P2 fails to produce a zero-fit match → the consensus dialect of §1.3 stays a dialect and this section is recorded as a negative, per the negatives-audit discipline. No partial credit; no post-hoc parameter rescue. Two-Madelung note for future sessions: the S5 receipt concerns the Madelung *rule* (aufbau ordering); the fluid mapping of §4 is the Madelung *transform*. Same Erwin Madelung, two different objects; the S5 receipt does not transfer.

8 · Sources (primary anchors)

- Onnes 1911; Meissner & Ochsenfeld 1933; London 1935; Ginzburg–Landau 1950; BCS, *Phys. Rev.* 108, 1175 (1957); Abrikosov 1957; Eliashberg 1960; Josephson 1962.
- Deaver & Fairbank; Doll & Näbauer, *PRL* 7 (1961) — flux quantum h/2e.
- Bednorz & Müller, *Z. Phys. B* 64, 189 (1986); Wu et al., *PRL* 58, 908 (1987) — YBCO.
- Uemura et al., *PRL* 62, 2317 (1989). McMillan, *Phys. Rev.* 167, 331 (1968); Allen & Dynes, *PRB* 12, 905 (1975).
- Troyer & Wiese, *PRL* 94, 170201 (2005) — sign problem NP-hardness.
- Duan et al., *Sci. Rep.* 4, 6968 (2014); Drozdov, Eremets et al., *Nature* 525, 73 (2015) — H_3S. Liu et al., *PNAS* 114, 6990 (2017); Peng et al. 2017; Drozdov et al., *Nature* 569, 528 (2019); Somayazulu et al., *PRL* 122, 027001 (2019) — LaH_{10}.
- Fausti et al., *Science* 331, 189 (2011); Mitrano et al., *Nature* 530, 461 (2016); Budden et al., *Nat. Phys.* 17, 611 (2021); Hu et al., *Nat. Mater.* 13, 705 (2014) — YBCO transient signatures to

300 K; Rowe et al., *Nat. Phys.* (2023) — resonant ×100 fluence reduction — driven superconductivity.

- Cao et al., *Nature* 556, 43 (2018) — magic-angle TBG. Kamiya et al., *Nat. Commun.* 9, 154 (2018) — quasicrystal superconductivity. Dolui et al., *PRL* 132 (2024) — Mg_2IrH_6 ambient-pressure ab initio prediction (unsynthesized).
- Ueno et al., *Nat. Mater.* 7, 855 (2008) — gate-induced SC in $SrTiO_3$; Ye et al., *Science* 338, 1193 (2012) — MoS_2 — the honest descendants of the electrostatic-doping premise (Schön audit).
- Gupta, arXiv:1007.2736 (2010) — unreplicated Al/PZT 313 K anomaly; provenance of the Pais wire design (audit Addendum 3); adjudicated by the §5.1 bench.
- Fan et al., *PRL* 130, 071801 (2023) — electron g-2. Saminadayar et al.; de-Picciotto et al. (1997) — e/3. Jompol et al., *Science* 325, 597 (2009) — spin–charge separation.
- In-repo: Pais audit (incl. Addendum 3, AIAA paper); Schön fulleride audit; phonon drive bandwidth analysis; scoring receipts `lpoh/sims/PREREG_SC_P1P3_2026-08-01.md` → `RESULTS_SC_P1P3_2026-08-01.md` and `PREREG_SC_P4_2026-08-01.md` → `RESULTS_SC_P4_2026-08-01.md`; S5 TF-consensus receipt (memory: project_scalar_channel_s4_2026-07-19).

Firewall: this is a research-lane document. Nothing here is deck material; the deck stays measured-receipts-only, and this paper's measured content (the table, §3, §5's Cavalleri receipts) is public-domain literature, not project receipts.